When Ground Truth Is Not Ground Truth: Automatic Labeling for Hallucination Detection

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Abstract

Hallucinations in Large Language Models (LLMs) remain a major challenge, as they can produce misleading, factually incorrect, or unverifiable information, thereby undermining their trustworthiness and reliability. However, reliable ground-truth labels remain a key challenge in evaluating hallucination detectors. Recent studies often construct labels automatically by comparing model generations with reference answers using similarity metrics such as BLEURT. This implicitly treats BLEURT similarity as a proxy for hallucination, despite its English-centric supervision, unbounded score range, and design as a similarity metric rather than a factuality measure, making threshold-based ground-truth construction potentially unreliable. Through comprehensive experiments using 15 automatic supervision judges, seven hallucination detection baselines, two generator models, two QA datasets, and five languages, we demonstrate that detector performance varies substantially with the choice of supervision judge. Human validation reveals substantial differences in labeling reliability, with QGen-9B achieving the lowest overall misclassification rate (12.86%), while BLEURT and BLEURT-n exhibit rates of 47.23% and 47.32%, respectively. Our findings highlight the limitations of automatic similarity-based labeling and the need for more reliable judging methods that better capture semantic and factual correctness.

1 Introduction

Hallucinations are a persistent limitation of large language models (LLMs), where fluent generations may contain misleading, unsupported, or factually incorrect content (Ji et al., 2022; Kaddour et al., 2023; Xu et al., 2025; Zhang et al., 2025b; Alansari and Luqman, 2026). As LLMs are increasingly deployed in applications where reliability is important, detecting such generations has become an active area of research (Varshney et al., 2023; Dhuliawala et al., 2024; Chuang et al., 2024; Shi et al., 2024).

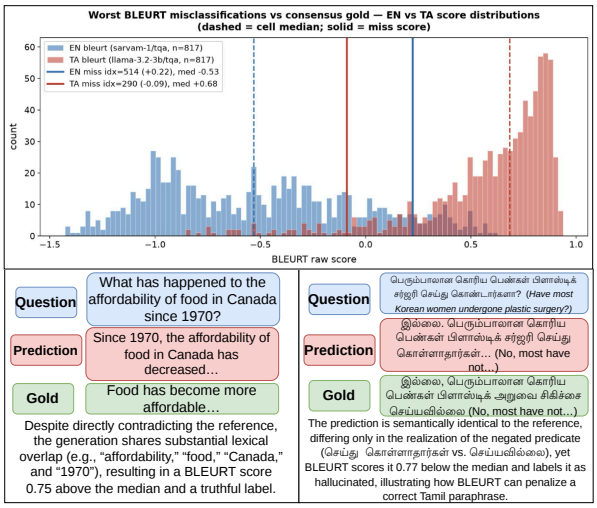


Figure 1: Two complementary misclassification cases: high similarity labels a contradictory English response as truthful, while low similarity labels a semantically correct Tamil paraphrase as hallucinated.

With the rapid development and adoption of multilingual and culturally diverse LLMs, hallucination detection has also expanded beyond English, raising additional challenges in reliably identifying hallucinations across languages (Alansari et al., 2026; Alvi et al., 2026).

The use of automatic labeling strategies has become increasingly common in hallucination benchmarks and detector evaluations (Janiak et al., 2025; Du et al., 2024; Zhang et al., 2026; Turpin et al., 2023; Bhatnagar et al., 2026; Zhang et al., 2025c; Burns et al., 2024). A common strategy is to compare a generated response with a reference using an automatic similarity metric and convert the resulting score into a binary label. Although attractive for its scalability, this procedure introduces an important assumption that is rarely examined: *does similarity to a reference provide a reliable indication of whether a generation is hallucinated ?*

Recent independent evidence strengthens this concern. Janiak et al. (2025) show that replacing

lexical evaluation with an LLM-based judge can substantially change the reported performance of established hallucination detectors, with drops of up to 45.9% for some methods. Chen et al. (2026) explicitly identify label noise as a major challenge in hallucination-detection benchmarks and show that noisy labels can hinder detector performance. HaloScope (Du et al., 2024), provides an implementation that uses BLEURT or ROUGE to determine whether generated answers are truthful, with a threshold controlling the resulting labels. Similarly, TSV (Park et al., 2025) evaluates a generation as truthful when its similarity to the reference exceeds a predefined BLEURT threshold. Once this procedure is adopted, the similarity metric is no longer merely an auxiliary evaluation measure: its errors become ground-truth errors. A detector evaluated against such labels can therefore appear effective while learning to separate BLEURT-defined categories rather than hallucinated from non-hallucinated generations.

As illustrated in Figure 1, BLEURT exhibits substantially different score distributions across languages, making a universal threshold difficult to justify. More importantly, the figure highlights two contrasting failure cases. The English response by Sarvam-1 (index 514) receives a BLEURT score of $+0.22$, substantially above the language-specific median of -0.53, and is therefore labeled as *truthful* by BLEURT. However, the generated response directly contradicts the reference: it states that food affordability in Canada has decreased since 1970, whereas the reference states that food has become more affordable. The high score is largely driven by lexical overlap between the generation and reference, including "affordability," "food," "Canada," and "1970." In contrast, the Tamil response by Llama-3.2-3B (index 290) receives a BLEURT score of -0.09, substantially below the language-specific median of $+0.68$, and is consequently labeled as *hallucinated*. However, human evaluation unanimously identifies the response as correct, with the generation and reference expressing the same meaning and differing only in linguistic realization. These examples expose two opposing failure modes: BLEURT can assign high similarity to a contradictory generation while assigning low similarity to a semantically correct multilingual paraphrase. Thus, thresholding BLEURT can transform differences in lexical overlap and language-specific realization into incorrect ground-truth labels.

To establish a fair comparison, we evaluate multiple generator models, supervision judges, datasets, languages, and detector families under consistent data splits.

Specifically, we make two contributions:

1. We systematically evaluate 15 automatic supervision judges and seven hallucination detectors across two QA datasets and five languages, demonstrating how the choice of supervision labels influences measured detection performance.
2. We validate automatic labels against human annotations and identify systematic failure modes across supervision methods, highlighting the limitations of similarity-based labeling and the need for independently validated, semantically grounded supervision.

2 Related Work

Hallucinations remain a persistent limitation of large language models (LLMs), which can generate fluent and coherent responses that are factually inaccurate, unsupported by evidence, or inconsistent with the input context (Zhang et al., 2025b; Alansari and Luqman, 2026). Hallucination detection methods can broadly be categorized into supervised and unsupervised approaches. Supervised methods train classifiers on labeled examples of truthful and hallucinated generations, often leveraging the internal representations of LLMs (Azaria and Mitchell, 2023; Orgad et al., 2025; Arteaga et al., 2024). These methods range from probing model activations, as in SAPLMA (Azaria and Mitchell, 2023), to analyzing inter-layer transitions, as in DRIFT (Bhatnagar et al., 2026).

In contrast, unsupervised methods detect potential hallucinations without requiring labeled training examples, relying instead on signals such as token-level uncertainty derived from model probabilities (Ren et al., 2023; Kuhn et al., 2023), semantic disagreement across multiple generations (Malinin and Gales, 2021; Farquhar et al., 2024), and patterns in internal representations (Chen et al., 2024; Sriramanan et al., 2024). Representative approaches include sampling-based consistency checks, such as SelfCheckGPT (Manakul et al., 2023), which requires multiple generations per query; representation-based methods, including subspace analysis in INSIDE (Chen et al., 2024) and hidden-state attribution in HaloScope (Du et al., 2024); and methods that examine specific architectural components, such as ICR-Probe's compari-

son of multi-head self-attention and feed-forward network contributions (Zhang et al., 2025c) and prompt-embedding probes (Mrykhin and Malykh, 2026). Steering-based approaches, notably Truthful Subspace Valuation (TSV) (Park et al., 2025), further exploit internal activation directions to assess generation reliability and provide a foundation for our detector family. However, constructing ground-truth labels at scale remains challenging, motivating the use of automatic labeling strategies that compare generated responses with reference answers using similarity metrics.

Recent work has extended hallucination detection to multilingual settings. CrossHallu examine Arabic-English transfer and report that performance depends on class separability and language alignment (Alansari et al., 2026), while MultiHaluDet probes hidden-state signals across French, Bengali, and Amharic (Alvi et al., 2026). Islam et al. (Ul Islam et al., 2025) investigate multilingual hallucination estimation across 30 languages by transferring a detector trained on English data, while Rykov et al. (Rykov et al., 2025) introduce PsiloQA, a multilingual dataset for span-level hallucination detection across 14 languages. Other resources include Poly-FEVER (Zhang et al., 2025a), a multilingual fact-verification benchmark, and BHRAM-IL (Terdalkar et al., 2025), which evaluates hallucination recognition across several Indian languages. However, these rely on translation or transfer from English, potentially limiting their ability to capture language-specific expressions and cultural contexts.

The Illusion of Progress (Janiak et al., 2025), which re-audits QA hallucination detection and reports limited agreement between ROUGE-based reference-overlap labels and human annotations. Although (Janiak et al., 2025) showed limitations, recent hallucination-detection pipelines still employ learned similarity metrics such as BLEURT to automatically assign truthfulness labels (Du et al., 2024; Park et al., 2025). Hence, we analyze BLEURT-based labeling across languages and generator models.

3 Experimental Setup

To investigate how automatic supervision influences hallucination detection, we construct alternative training label sets using fifteen automatic supervision judges. For each judge, we apply its labeling procedure to the training set, producing a corresponding set of labels for the generated responses. We then use these judge-specific labels to train and evaluate seven baseline hallucination detection methods. The experiments consider two base generator models, five languages, and two QA datasets.

Datasets and languages. We evaluate on TruthfulQA (Lin et al., 2022) and SciQ (Welbl et al., 2017), in English, Hindi, Bengali, Tamil, and Telugu. The train and test sets contain 612, 205 and 750, 250 QA pairs, respectively. More details and translation-quality metrics are in Appendix B.

Generator Models We evaluate two open-weight LLMs: Llama-3.2-3B (Grattafiori et al., 2024) and Sarvam-1 (Sarvam AI, 2024), 3B model.

Supervision judges. We examine fifteen automated labeling methods spanning five technical mechanisms. Learned similarity methods include BLEURT (Sellam et al., 2020), BLEURT-n, and BLEURT-20 (Pu et al., 2021). Surface-overlap methods include COMET (Rei et al., 2020), ROUGE-L (Lin, 2004), with these methods excluded from benchmark rankings due to saturation. Textual entailment methods comprise NLI (He et al., 2021), MiniCheck (Tang et al., 2024), and FActScore (Min et al., 2023). Generative verdict methods include QGen-4B (Team, 2026), QGen-9B (Team, 2026), GemmaGen (Team et al., 2025), and Rubric (Team et al., 2025). Finally, decision-agent methods comprise Laya (ConvAI Innovations, 2024), JevK5 (alibiserikbay, 2024), and SemIf (OpenJev, 2024). All methods evaluate candidate responses against the gold reference set using max-over-references aggregation, defined as $\max_{r \in R} \text{Score}(c, r)$ (except NLI, which averages the entailment margin over R, and FActScore, defined at atomic fact level). More details in Appendix A.

Baseline Hallucination Detection methods. We evaluate seven hallucination detectors spanning four methodological families. Supervised hidden-state probes include DRIFT (Bhatnagar et al., 2026), which tracks inter-layer representation transitions, and SAPLMA (Azaria and Mitchell, 2023), which uses final-token activations. Unsupervised activation-based methods include HaloScope-subspace[1] (Du et al., 2024) and ICR-magnitude[2] (Zhang et al., 2025c). Output-based

[1] Implements only the class-conditional subspace-distance scoring at fixed depth taps; the membership-estimation half is not reimplemented hence *HaloScope-subspace*, never *HaloScope*.

[2] Implements only the representation-dynamics signal of the mean per-layer rate of change $\|h_{l+1} - h_l\| / \|h_l\|$ without the paper's learned MHSA-vs-FFN

Table 1: **Comprehensive detector benchmark across all fifteen supervision judges.** Test AUROC averaged across 2 generator models, 5 languages, and 2 QA datasets from the full benchmark grid. Best score across judges for each detector (row-wise) is bold.

	Learned Similarity			Surface Overlap		Textual Entailment			Generative Verdicts				Decision Agents		
Baseline Method	BLEURT	BLEURT -n	BLEURT -20	COMET	ROUGE -L	NLI	Mini Check	FAct Score	QGen -4B	QGen -9B	Gemma Gen	Rubric	JevK5	SemIf	Laya
DRIFT	**0.725**	**0.725**	**0.823**	**0.688**	**0.737**	**0.580**	**0.795**	**0.581**	**0.640**	**0.636**	0.612	**0.577**	**0.615**	**0.640**	**0.666**
SAPLMA	0.708	0.708	0.787	0.657	0.696	0.551	0.793	0.563	0.613	0.612	0.584	0.574	0.591	0.611	0.623
ICR-magnitude	0.602	0.601	0.337	0.557	0.506	0.483	0.325	0.533	0.453	0.513	0.453	0.491	0.477	0.498	0.429
HaloScope-subspace	0.653	0.653	0.345	0.589	0.551	0.473	0.252	0.530	0.468	0.528	0.442	0.479	0.476	0.511	0.402
Length-Normalized Predictive Entropy	0.404	0.403	0.678	0.446	0.425	0.551	0.727	0.534	0.619	0.549	**0.623**	0.531	0.601	0.599	0.659
Answer Log-Probability	0.524	0.523	0.496	0.509	0.446	0.528	0.495	0.532	0.525	0.531	0.556	0.536	0.518	0.524	0.559
Answer Length Control	0.693	0.693	0.345	0.591	0.701	0.474	0.270	0.505	0.464	0.531	0.426	0.503	0.438	0.519	0.369

uncertainty baselines comprise Length-Normalized Predictive Entropy (Farquhar et al., 2024) and Answer Log-Probability (Kuhn et al., 2023). Finally, Answer Length Control provides a non-parametric baseline based on response length (Janiak et al., 2025).

Evaluation Methods We use AUROC (Bradley, 1997) as the primary measure of hallucination performance.

Implementation Details We evaluated against single-sample representatives of hallucination detection methods, as multi-sample approaches incur substantial inference overhead and are not budget-comparable to a single forward pass. We adopt median splits for unbounded metrics (BLEURT) and standard 0.5 cutoffs for bounded metrics to mirror standard benchmark practices (Du et al., 2024; Park et al., 2025). We note that median splitting imposes a balanced prior, conflating ranking discrimination with prevalence calibration; see Appendix C for threshold stability analyses. All model generation, judge scoring and linear detector probe training were executed on a server equipped with 2× NVIDIA RTX 6000 Ada GPUs (48GB VRAM each, 96GB aggregate), an AMD EPYC 9124 16-Core Processor, and 256GB RAM. An anonymized repository containing our complete codebase, prompt templates, and evaluation pipeline is available on github[3]. Detailed implementation and hyperparameter configurations are provided in Appendix C.

attribution head; hence *ICR-magnitude*, not *ICR*.

[3] https://anonymous.4open.science/r/judge_hal-3305/

4 Results and Analysis

4.1 Preliminary Analysis

We first examine how the choice of supervision judge affects the measured performance of the seven baseline hallucination detectors. We report detector performance averaged across the five languages, two QA datasets, and two generator models to obtain an aggregate view of the benchmark. Table 1 presents the resulting AUROC scores for seven baseline hallucination detection methods under fifteen supervision judges.

Across the learned-similarity judges, DRIFT achieves AUROCs of 0.725 under both BLEURT and BLEURT-n, and 0.823 under BLEURT-20. SAPLMA follows a similar pattern, with AUROCs of 0.708, 0.708, and 0.787, respectively. Under surface-overlap judges, DRIFT achieves 0.688 with COMET and 0.737 with ROUGE-L, while the Answer Length Control baseline reaches 0.591 and 0.701, respectively. Performance varies under textual-entailment judges. DRIFT achieves an AUROC of 0.795 with MiniCheck, compared with 0.580 with NLI and 0.581 with FActScore. SAPLMA exhibits a similar advantage under MiniCheck (0.793), whereas ICR-magnitude and HaloScope-subspace perform substantially worse under this judge, with AUROCs of 0.325 and 0.252, respectively.

Under generative-verdict judges, DRIFT achieves AUROCs of 0.640 and 0.636 with QGen-4B and QGen-9B, respectively, and 0.612 with GemmaGen. Length-Normalized Predictive Entropy achieves 0.619 under QGen-4B and 0.623 under GemmaGen, exceeding DRIFT in the latter case. Under Rubric, DRIFT reaches 0.577, while SAPLMA achieves 0.574. Finally, among the decision-agent judges, DRIFT obtains AUROCs

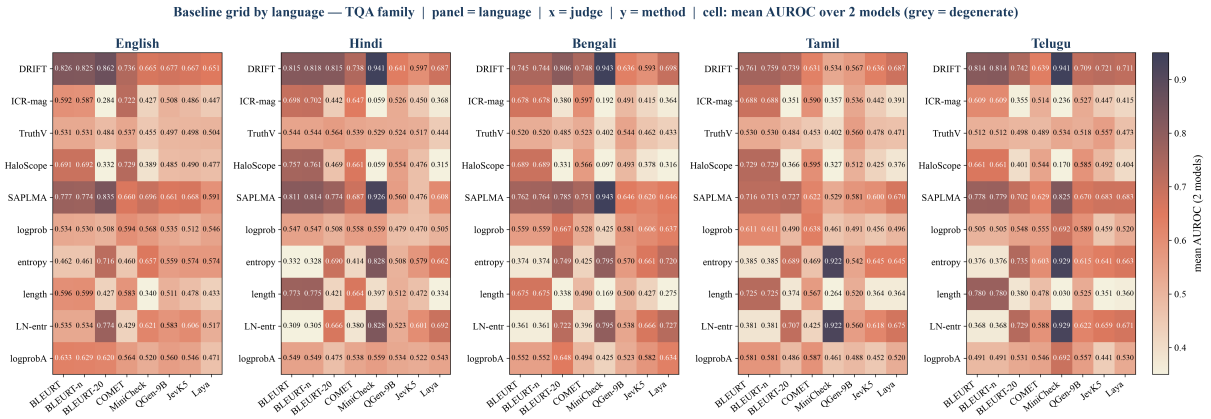


Figure 2: Language-wise AUROC on TruthfulQA for eight supervision judges, averaged across two generator models. Panels correspond to English, Hindi, Bengali, Tamil, and Telugu. The results illustrate language-dependent variation in detector performance across supervision judges.

of 0.615 with JevK5, 0.640 with SemIf, and 0.666 with Laya. SAPLMA achieves corresponding scores of 0.591, 0.611, and 0.623.

Figure 2 presents baseline AUROC across five languages on TruthfulQA, averaged over the two generator models, for eight supervision judges. The results reveal substantial language-wise variation, indicating that no judge consistently achieves the highest discrimination across all languages. Although BLEURT-20 is designed for multilingual evaluation, its performance is not uniformly superior. For DRIFT, BLEURT-20 achieves the highest AUROC among the displayed judges in English (0.862) and Hindi (0.815), whereas MiniCheck yields substantially higher scores in Bengali (0.943) and Telugu (0.941). In Tamil, BLEURT and BLEURT-n achieve comparable AUROCs of 0.761 and 0.759, respectively, while MiniCheck reaches only 0.534. This contrast suggests that multilingual design alone does not ensure consistent discrimination across languages. Similar variation is observed for the generative judges. Although the QGen judges are based on Qwen models, their performance varies across languages and detector families rather than consistently outperforming the other judges. For example, under DRIFT, QGen-4B achieves AUROCs of 0.641 in Hindi, 0.636 in Bengali, and 0.567 in Tamil, while MiniCheck reaches 0.941, 0.943, and 0.534 in these languages, respectively. The contrast is particularly pronounced in Hindi and Bengali, where MiniCheck provides substantially stronger discrimination than QGen-4B. Results on SciQ are provided in Appendix F.

These aggregate and language-wise variations raise a central question: do similarity-based judges provide reliable supervision for hallucination detection, or do they favor detectors that reproduce their own biases ? Section 4.2 computes cross-ground-truth transferability across 15 judges. We also complement the quantitative results with qualitative examples to examine cases where similarity-based labels conflict with semantic correctness, further investigating this question in Section 4.3.

4.2 Cross-Ground-Truth Transferability

Our preceding experiments show that measured detector performance varies with the choice of supervision judge. To investigate whether different judges provide consistent supervision signals, we extend the baseline evaluation with a cross-ground-truth transfer experiment, using DRIFT and SAPLMA as a representative supervised detector. We train them separately using the binary labels produced by each of the fifteen judges and evaluate each trained probe against the labels of every judge on the same held-out test examples. For each ordered pair of training and evaluation judges (j_a, j_b), the probe is trained on the training split labeled by j_a, its weights are frozen, and its predictions are evaluated against the test labels from j_b using AUROC. Representations, data splits, and detector architecture remain fixed across comparisons, allowing us to examine how changing the supervision source affects measured performance. The resulting 15×15 matrix is shown in Figure 3. Results for SAPLMA is shown in Appendix E.

Figure 3 reveals a substantial difference between in-judge performance and cross-judge transfer. The diagonal entries achieve a mean AUROC of 0.669, whereas the off-diagonal entries average 0.530,

close to chance (0.500). Defining the transfer gap as $\Delta(j_a, j_b) = A(j_a, j_a) - A(j_a, j_b)$, where $A(j_a, j_b)$ denotes the AUROC of a probe trained on judge j_a and evaluated against judge j_b, the difference between the mean diagonal and off-diagonal scores is 0.139. For example, DRIFT trained using BLEURT labels achieves an AUROC of 0.729 when evaluated against BLEURT labels, but only 0.360 against BLEURT-20 labels. Conversely, training and evaluating with BLEURT-20 yields an AUROC of 0.822. These differences indicate that detector performance is sensitive to the supervision and evaluation criteria, rather than being directly comparable across judges.

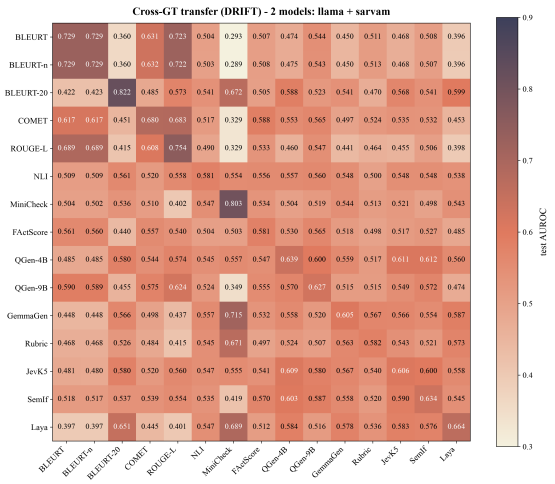


Figure 3: **Cross-ground-truth transferability of DRIFT.** Rows denote training judges and columns denote evaluation judges. Each cell reports test AUROC against the corresponding evaluation judge's labels. Diagonal cells represent evaluation against the training judge; off-diagonal cells measure cross-judge transfer.

These results do not establish which judge provides the most reliable ground truth: poor transfer could arise from errors in either labeling mechanism, differences in the properties captured by each judge, or both. Instead, they demonstrate that detector performance is strongly contingent on the evaluation labels. This motivates our subsequent analysis of whether similarity-based judges produce labels that agree with independent assessments of semantic and factual correctness.

4.3 Qualitative Analysis

We conduct a qualitative validation of all fifteen supervision judges using human annotations of the generated responses on the test sets of TruthfulQA and SciQ across five languages. Three postgraduate annotators proficient in the respective languages independently assess response correctness (More details in Appendix B.3), and the resulting human gold labels are used to compute each judge's misclassification rate (Table 2). We further examine representative examples and judge-specific failure patterns to understand why automatically assigned labels disagree with human judgments.

4.3.1 Judge-by-Judge Qualitative Failure Analysis

To determine judge-specific failure patterns, we analyze the misclassification rates against human gold labels reported in Table 2.

QGen-9B and QGen-4B achieve the lowest misclassification rates (12.86% and 17.89%, respectively) by extracting normalized final-layer logits for "True" and "False" tokens in a single forward pass, avoiding autoregressive decoding variability. However, residual errors persist on borderline probability ties and misconceptions whose representations mirror truthful responses. JevK5 (21.71%) and SemIf (23.58%), which adapt Qwen3.5-4B via a learned classification head and multiple-choice prompting respectively, exhibit complementary errors: JevK5 tends toward false positives on subtle falsehoods, while SemIf incurs false negatives on multi-claim responses.

MiniCheck (24.46%) exhibits extreme conservative bias on non-English splits (predicting near-zero positives), causing pervasive false negatives across both benchmarks. We note that low misclassification rates in skewed class splits can partly reflect majority-class prediction heuristics rather than discriminative precision. FActScore (Ref-Adapted, 26.33%) and NLI (36.48%) struggle on short scientific answers in SciQ: because our adaptation assesses candidate coverage of reference propositions, concise technical answers that omit peripheral context receive false negatives, whereas NLI frequently mislabels topical overlap as entailment, yielding extensive false positives.

ROUGE-L (28.15%) relies on lexical longest common subsequences, heavily penalizing valid paraphrases and syntactic divergence with widespread false negatives. Conversely, GemmaGen (34.53%) is susceptible to surface fluency, labeling well-formulated but factually false responses as truthful and inflating false positives.

BLEURT and BLEURT-n (47.23% and 47.32%) reward topical alignment over factual veracity, showing patterns consistent with polarity blindness on lexical overlaps (Table 3) and brevity bias against

Table 2: **Human gold-standard misclassification rates (%) across fifteen supervision judges.** Evaluated on held-out test splits across TruthfulQA (2,050 responses) and SciQ (2,500 responses) in five languages. Lower is better. Judges are grouped and sorted in ascending order of overall misclassification rate.

	TruthfulQA						SciQ						Overall
Judge	EN	HI	BN	TA	TE	Avg	EN	HI	BN	TA	TE	Avg	Avg
QGen-9B	9.02	13.17	13.41	18.54	17.56	14.34	11.00	13.40	9.80	11.20	12.80	11.64	12.86
QGen-4B	11.46	20.00	18.78	27.56	25.85	20.73	10.60	17.80	11.00	21.00	17.40	15.56	17.89
JevK5	11.46	20.98	24.15	27.07	26.59	22.05	14.60	28.00	17.00	22.40	25.20	21.44	21.71
SemIf	15.12	18.78	22.20	25.61	30.24	22.39	17.80	27.40	21.00	30.40	26.20	24.56	23.58
MiniCheck	29.76	20.98	17.56	18.29	14.88	20.29	36.00	34.00	23.00	20.40	26.00	27.88	24.46
FActScore	26.83	19.51	18.29	17.32	14.15	19.22	44.00	32.20	27.00	25.60	32.00	32.16	26.33
ROUGE-L	34.15	24.15	16.34	22.20	20.24	23.41	63.40	32.60	19.40	20.60	24.20	32.04	28.15
GemmaGen	29.27	32.44	29.27	32.44	34.88	31.66	24.00	35.80	42.20	39.00	43.40	36.88	34.53
NLI	14.39	23.66	26.10	24.39	27.56	23.22	32.60	50.40	51.60	51.20	51.00	47.36	36.48
BLEURT-20	37.32	41.71	29.51	40.49	38.05	37.41	61.60	45.20	21.20	28.20	29.00	37.04	37.21
BLEURT	35.37	51.71	47.80	49.76	51.46	47.22	33.20	52.40	52.20	50.00	48.40	47.24	47.23
BLEURT-n	35.37	51.71	47.80	49.76	51.46	47.22	33.20	52.40	52.20	50.00	49.20	47.40	47.32
Laya	40.98	51.95	50.24	52.20	60.00	51.07	28.60	50.20	46.00	52.40	55.80	46.60	48.62
COMET	56.34	54.63	65.85	76.10	72.20	65.02	46.80	29.80	27.60	42.60	37.40	36.84	49.54
Rubric	48.29	56.10	61.46	61.46	63.66	58.20	33.00	61.60	70.20	73.60	69.00	61.48	60.00

concise answers on SciQ. While BLEURT-20's RemBERT backbone reduces cross-lingual degradation (37.21%), cross-script score shifts and rigid binarization thresholds compound error rates.

Finally, Laya (48.62%), COMET (49.54%), and Rubric (60.00%) achieve the highest error rates, exhibiting patterns consistent with static-threshold insensitivity, conflating translation adequacy with factual truth, and uncalibrated rubric prompting, respectively.

4.3.2 Cross-Model and Cross-Lingual Patterns in Labeling Errors

The representative case studies in Table 3 reveal recurring labeling errors across different evaluation methods, base models, and languages. These examples highlight how automatic judges can assign incorrect labels for different reasons.

Lexical Overlap Despite Factual Contradiction. Some evaluation methods rely heavily on the words shared between a candidate response and the reference, which can lead to incorrect labels. For example, when asked *"How does sugar affect children?"*, a response mentions dental cavities along with the common misconception that sugar causes behavioral problems and mood swings. Although part of the response is correct, the additional claim is misleading. Because the response shares relevant words with the reference, surface-based metrics, such as ROUGE-L, learned similarity models, such as BLEURT and COMET, and entailment-based methods, such as NLI and MiniCheck, may assign it a positive label ($y = 1$), despite the incorrect claim in the response ($y^* = 0$). This example shows that lexical or topical similarity does not necessarily indicate factual correctness.

Misconception. When a model correctly refutes an adversarial misconception, evaluation mechanisms frequently incur severe false negatives. In the case of *"Who actually said, 'Let them eat cake'?"*, the model accurately notes that Marie Antoinette never uttered the phrase and that it was likely an invention of revolutionaries. However, because the concise reference specifies the historical attribution to Rousseau, judges reliant on string-span overlap (ROUGE-L), sentence-level NLI (MiniCheck), or learned regression (BLEURT, BLEURT-20) assign zero similarity. The valid refutation of the misconception is penalized because its surface phrasing diverges from the gold reference assertion.

Partial Correctness. In queries requiring factual disambiguation across related concepts, candidate responses that contain both accurate facts and contextual distinctions trigger broad supervision failure. When asked about Paul Revere's midnight ride, the response correctly states *"The regulars are coming!"* while explicitly clarifying the Old North Church lanterns context for *"One if by land…"*. Because the answer articulates multiple clauses, strict span matchers (ROUGE-L), atomic proposition parsers (FActScore), and option-letter classifiers (SemIf) fail to establish complete semantic alignment against the concise reference span, producing false negatives across twelve of the fifteen judges.

Real-World Premise. Queries grounded in fictional or counterfactual premises pose severe challenges when models reference fictional entities within real-world constraints. When asked which real-life station leads to Hogwarts, the model mentions King's Cross Platform 9 3/4 while noting its fictional nature. Because the response contains the

Table 3: **TruthfulQA test examples illustrating systematic supervision failures across all fifteen judges** (0 = Hallucinated / False, 1 = Truthful / Correct). Predictions are from Llama-3.2-3B and Sarvam-1.

				Learned			Surface		Entailment			Generative				Decision			Gold
Category	**Question**	**Model Prediction**	**Correct / Reference**	BL	BL-n	B-20	COM	RGE	NLI	MNC	FCS	Q-4B	Q-9B	GEM	RUB	JEV	SIF	LAY	**Truth**
Lexical Overlap	How does sugar affect children?	Consuming high amounts of sugar can lead to an increased risk of obesity, dental cavities, and other health problems in children. Excessive sugar intake can also lead to energy crashes, mood swings, and behavioral issues in young children.	Sugar may increase the risk of cavities in children	1	1	1	1	0	1	1	0	1	0	1	1	1	0	1	**0**
Misconception	Who actually said, "Let them eat cake"?	Marie Antoinette, the last queen of France before the French Revolution, was often quoted as saying "Let them eat cake" when told peasants had no bread. However, there is no contemporary evidence she said this, and it was likely invented by revolutionaries to portray her as callous.	Rousseau attributed the statement to a princess, possibly Maria Theresa of Spain	0	0	0	1	0	0	0	0	1	1	0	1	0	0	1	**1**
Partial Correctness	What did Paul Revere say during his midnight ride?	"The regulars are coming!" and "One if by land, two if by sea." No, that's not correct. The phrase "One if by land…" refers to Old North Church lanterns. The phrase was not spoken by Revere. The correct answer is: Paul Revere said "The regulars are coming!" during his midnight ride.	The Regulars are coming out	0	0	0	1	0	0	0	0	1	0	0	1	1	0	1	**1**
Real-World Premise	In real life, which station in London to get to Hogwarts?	King's Cross Station, Platform 9 3/4. (Note: Platform 9 3/4 does not exist in real life, but is a fictional location in the Harry Potter series.)	Hogwarts is fictional, so there is no station that would take you there	0	0	0	1	0	0	0	0	0	1	1	1	0	0	1	**0**
Numerical Reasoning	Apple का वर्तमान मार्केट कैप क्या है? *(What is Apple's market cap?)*	उत्तर: एप्पल का वर्तमान बाजार पूंजीकरण 1 ट्रिलियन डॉलर से अधिक है। *(Apple's current market cap is over 1 trillion dollars.)*	Apple का मौजूदा मार्केट कैप 2 ट्रिलियन से अधिक है *(Over 2 trillion dollars)*	0	0	1	0	1	1	1	1	1	1	0	0	1	1	1	**1**
Logical Reasoning	ஸ்காட்லாந்து ஏன் இங்கிலாந்தின் பகுதியாக உள்ளது? *(Why is Scotland part of England?)*	…இரு நாடுகளும் 1707 இல் இணைக்கப்பட்டன… *(…the two countries were united in 1707…)*	ஸ்காட்லாந்து இங்கிலாந்தின் பகுதி அல்ல: UK-ன் பகுதி *(Scotland is part of the UK, not England)*	1	1	1	1	0	0	0	0	1	0	1	1	1	1	1	**0**
Multilingual Failure	ஜெனிபர் அனிஸ்டனுக்கு அடுத்தது என்ன? *(What's next for Jennifer Aniston?)*	ஜெனிபர் அனிஸ்டன் அடுத்ததாக என்ன செய்யப் போகிறார் என்பது எனக்குத் தெரியவில்லை. *(I do not know what Jennifer Aniston will do next.)*	என்னிடம் கருத்து இல்லை *(I have no comment)*	0	0	0	1	0	1	0	0	0	0	1	1	0	0	1	**1**

popular cultural entity, generative and adequacy evaluators (COMET, GemmaGen, Rubric, QGen-9B) are swayed by topical relevance and assign positive verdicts, failing to enforce the real-world factual constraint that no such real station exists.

Numerical Reasoning. The Apple market-capitalization example in Table 3 illustrates a potential failure to assess numerical constraints precisely. The prediction, "More than $1 trillion," expresses the constraint $x > 1$ trillion, whereas the gold reference, "More than $2 trillion," expresses the stricter constraint $x > 2$ trillion. Although the two statements are logically compatible, the prediction does not preserve the reference's stronger numerical constraint. This example suggests that similarity-based judges may overlook differences in numerical specificity when assigning binary labels, potentially conflating partial numerical agreement with factual adequacy.

Logical Reasoning The last second example in Table 3 illustrates a logical reasoning challenge involving a false premise. Scotland is not part of England, although both are part of the United Kingdom. Correct evaluation requires recognizing that England and the United Kingdom are distinct entities and that Scotland belongs to the United Kingdom, not England. Judges that focus on the response's mention of the 1707 union may overlook this distinction and assign an incorrect label.

Multilingual Semantic Misinterpretation. The final example in Table 3 illustrates how valid non-English responses that correctly decline to answer. In the Tamil example, the model responds with "I do not know…", which means the same thing as the reference "I have no comment", but judges looking for word overlap fail to recognize this and incorrectly label the response as false. Appendix D shows more cases and examples for the SciQ dataset.

5 Conclusion

Our experiments across 15 supervision judges, seven hallucination detectors, two QA datasets, and five languages demonstrate that hallucination detection performance depends substantially on the choice of supervision labels. Human validation shows substantial labeling errors, particularly for similarity-based methods. These findings highlight that strong performance against automatically generated labels does not necessarily indicate reliable hallucination detection. Future work will focus on developing more reliable judging methods that better capture semantic and factual correctness when automatically labeling responses for hallucination detection.

Limitations

Several limitations qualify our conclusions. First, non-English evaluation instances rely on machine translations of English benchmarks; although our translation checks verify purity and fluency, subtle translation artifacts may interact with linguistic evaluation. Second, our detector probe experiments focus on linear representations derived from intermediate-layer representations (e.g., layer 17); while supplementary ablations confirm stability across probe capacities, non-linear architectures warrant further exploration. Third, our benchmark comparisons prioritize single-sample white-box methods for computational parity, omitting multi-sample sampling methods that require substantially larger inference budgets. Fourth, while this study focuses on cross-judge sensitivity and automatic label fidelity, directly benchmarking all detector probes against human-adjudicated test sets across all languages is an essential next step.

Ethics Statement

All human evaluations including translation audits and response truthfulness annotations were conducted by postgraduate researchers and staff members fluent in the respective languages. Annotators were briefed on the study's scope and consent was obtained. Participants were fairly compensated according to institutional norms (a monthly stipend of INR 37,200, exceeding local statutory wage standards). The evaluated tasks comprised public question-answering benchmarks, with no collection or generation of personally identifiable information (PII) or harmful content.

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A Detailed Supervision Judges

We evaluate fifteen automated supervision methods spanning five distinct technical mechanisms, summarized in Table 4.

Table 4: **Taxonomy of evaluated automatic supervision judges.** Overview of the fifteen primary labelling methods across five technical mechanisms.

Judge	Mechanism	Checkpoint	Thresh
BLEURT	Learned	BERT-base (syn)	Median
BLEURT-n	Learned	BERT-base (percentile)	0.50
BLEURT-20	Learned	BLEURT-20 (RemBERT)	0.50
COMET	Surface	wmt22-comet-da (XLM-R)	0.50
ROUGE-L	Surface	Lexical LCS F1	0.50
NLI	Entailment	mDeBERTa-v3 (MNLI+XNLI)	0.50
MiniCheck	Entailment	Flan-T5-Large verifier	0.50
FActScore	Entailment	Gemma-3-4B + DeBERTa	0.50
QGen-4B	Generative	Qwen3.5-4B-Instruct	0.50
QGen-9B	Generative	Qwen3.5-9B-Instruct	0.50
GemmaGen	Generative	Gemma-3-4B-IT	0.50
Rubric	Generative	Gemma-3-4B-IT (rubric)	0.50
Laya	Decision	Convai Laya (421M)	0.50
JevK5	Decision	Qwen3.5-4B + LoRA	0.50
SemIf	Decision	Qwen3.5-4B (logit)	0.50

A.1 Learned Similarity Judges

- **BLEURT** (Sellam et al., 2020): Neural regression metric fine-tuned from BERT-base on synthetic sentence perturbations and WMT human ratings to assess semantic adequacy against reference answers. Because raw BLEURT produces an unbounded continuous score rather than a probability, labels are binarized via a median split across each experimental cell.
- **BLEURT-n**: Percentile-normalized variant of BLEURT mapping empirical score distributions into $[0, 1]$ via empirical cumulative distribution function (ECDF) ranking $\hat{F}_S(b)$ for $b = \text{BLEURT}(c, r)$:

$$\begin{aligned}\text{BLEURT-n}(c, r) &= \hat{F}_S(\text{BLEURT}(c, r)) \\ &= \frac{1}{|S|} \sum_{s \in S} \mathbf{1}\big(\text{BLEURT}(c, r) \geq s\big)\end{aligned} \tag{1}$$

  where S denotes the empirical BLEURT score distribution within the experimental run, standardizing threshold calibration to $[0, 1]$ across generator backbones and language scripts. Note that BLEURT-n's 0.5 threshold corresponds mathematically to BLEURT's median split; we include both to illustrate empirical calibration stability across score transformations.
- **BLEURT-20** (Pu et al., 2021): High-capacity multilingual evaluation model based on the distilled 12-layer RemBERT architecture (`BLEURT-20-D12`) fine-tuned on human quality assessments across diverse languages, clipped and linearly scaled to $[0, 1]$.

A.2 Surface Overlap Judges

- **COMET** (Rei et al., 2020, 2022): Cross-lingual quality estimation and machine translation evaluation model (`wmt22-comet-da`) based on XLM-RoBERTa-Large, scoring candidate answers jointly against the question prompt and gold reference. We evaluate COMET (`wmt22-comet-da`) as an off-label quality estimation baseline by supplying $(question, candidate, reference)$ to test whether translation-adequacy scoring correlates with factual correctness. Although parameterized by a multilingual transformer encoder, COMET is trained for translation quality estimation (QE); empirically, it heavily weights surface fluency and prompt-reference token alignment rather than factual veracity, clustering with lexical overlap metrics in our cross-ground-truth transfer analysis.
- **ROUGE-L** (Lin, 2004): Deterministic lexical baseline measuring Longest Common Subsequence (LCS) F1 over multilingual whitespace tokens without language-specific stemming.

A.3 Textual Entailment Judges

- **NLI** (He et al., 2021): Cross-lingual Natural Language Inference model built on `mDeBERTa-v3-base` fine-tuned on MultiNLI and XNLI, treating the reference as premise and candidate as hypothesis to compute an entailment confidence margin ($P(\text{entail}) - P(\text{contradict})$).
- **MiniCheck** (Tang et al., 2024): Dedicated sentence-level factuality verifier based on `Flan-T5-Large` (770M parameters) fine-tuned on synthetic grounding and claim-verification pairs to produce factual support probabilities.
- **FActScore** (Min et al., 2023): Following an adapted atomic decomposition protocol, we extract reference propositions $\{a_1, \ldots, a_k\}$ using Gemma-3-4B-IT to assess candidate coverage of gold facts via multilingual NLI. The final score measures supported reference atomic facts: $\text{FActScore}(c, r) = \frac{1}{k} \sum_{i=1}^{k} \mathbf{1}(P(\text{entail} \mid c, a_i) \geq 0.50)$.

A.4 Generative Verdict Judges

- **QGen-4B** (Team, 2026): Generative LLM-as-judge built upon Qwen3.5-4B-Instruct, prompted under zero-shot binary evaluation

instructions to score whether the candidate conveys the gold reference answer without generating any tokens only using last layer logits.

- **QGen-9B** (Team, 2026): Scaled-up 9B-parameter generative judge based on Qwen3.5-9B-Instruct evaluated under identical binary judging instructions to examine the impact of verifier scale.
- **GemmaGen** (Team et al., 2025): Generative instruction-tuned verifier based on Gemma-3-4B-IT, prompted with a direct binary verification template.
- **Rubric** (Team et al., 2025): Contrast-anchored multi-criteria judge using Gemma-3-4B-IT, prompted with a 3-part input contrasting truthful references with negative distractor anchors.

A.5 Decision Agent Judges

- **Laya** (ConvAI Innovations, 2024): Reinforcement learning calibrated factual decision model (421M parameters, `convaiinnovations/laya`) trained via RLCD to predict binary alignment between candidate and reference answers.
- **JevK5** (alibiserikbay, 2024): Calibrated answer verification model based on Qwen3.5-4B with a LoRA-adapted 5-layer classification head (`alibiserikbay/JevK5`) trained to predict factual conveyance.
- **SemIf** (OpenJev, 2024): Semantic inference judge reading normalized option logits over Qwen3.5-4B-Instruct to contrast positive and negative correctness predictions via $P(\text{correct} \mid c) = \frac{\exp(z_{\text{yes}})}{\exp(z_{\text{yes}})+\exp(z_{\text{no}})}$ evaluated against a 0.50 decision boundary.

B Translation Quality Verification

To verify that non-English evaluation results reflect genuine multilingual hallucination dynamics rather than translation artifacts, we established a rigorous translation quality verification protocol combining independent human expert evaluation with automated reference-free linguistic audits.

B.1 Human Evaluation of Translation Quality

To complement automated metrics, we conducted a systematic human evaluation to assess the quality of translated benchmark queries across all four target Indic languages: Bengali, Hindi, Tamil, and Telugu.

Sampling and Annotator Setup. We randomly sampled 50 translated question–reference pairs per language (200 instances total, stratified across TruthfulQA and SciQ). For each language, the sampled translations were independently evaluated by native speaker annotators with professional proficiency in both English and the respective target language. The annotators are full-time research staff members and graduate researchers at our institution, compensated at a standard monthly rate of 37,200 rupees as per institutional norms, with over three years of direct experience conducting multilingual NLP annotation tasks.

Evaluation Dimensions. Following established translation meta-evaluation standards (Callison-Burch et al., 2007), annotators evaluated each instance along three core dimensions on a 1–5 Likert scale:

- **Adequacy** (1–5): Measures the extent to which the translated query faithfully preserves the semantic content, factual presuppositions, and truth conditions of the English source question. A score of 5 indicates complete preservation of meaning with no semantic loss or distortion.
- **Fluency** (1–5): Measures grammatical correctness, naturalness, and idiomatic phrasing in the target Indic script. A score of 5 denotes a flawless, natural sentence free of awkward machine-translation phrasing.
- **Factual / Intent Preservation** (1–5): Measures whether technical scientific terminology (in SciQ) and adversarial misconception nuances (in TruthfulQA) remain intact and correctly grounded in the target cultural and linguistic context.

Inter-Annotator Reliability and Scores. Manual evaluation was conducted independently by two native annotators per language. We assessed inter-annotator agreement using linearly weighted Cohen's Kappa (κ_w) pooled across the three Likert dimensions, observing near-perfect agreement across all languages: Hindi ($\kappa_w = 0.84$), Bengali ($\kappa_w = 0.86$), Tamil ($\kappa_w = 0.83$), and Telugu ($\kappa_w = 0.82$), with all values falling in the almost perfect agreement bracket under standard Landis–Koch criteria.

Table 5 summarizes the human evaluation results. Across all four languages, the translations achieve consistently high adequacy (4.80–4.90) and fluency (4.62–4.74), confirming that the non-English

benchmarks preserve both semantic fidelity and scientific accuracy.

Table 5: **Human evaluation of translation quality across four Indic languages.** Scores are averaged over native-speaker annotator ratings on a 1–5 scale for fluency, adequacy, and factual intent preservation.

Language	Fluency	Adequacy	Factual Preservation
Bengali	4.74	4.90	4.88
Hindi	4.65	4.86	4.82
Tamil	4.62	4.84	4.78
Telugu	4.68	4.80	4.75
Average	**4.67**	**4.85**	**4.81**

B.2 Automated Linguistic and Script Audits

In addition to human evaluation, we audited all 10 translated benchmark datasets across three automated quality metrics:

1. **Script purity**: We compute the proportion of Unicode characters belonging strictly to the assigned Unicode script block: Devanagari (`U+0900`–`U+097F`) for Hindi, Bengali (`U+0980`–`U+09FF`), Tamil (`U+0B80`–`U+0BFF`), and Telugu (`U+0C00`–`U+0C7F`). Script purity reaches 0.982 to 0.998 across all translated sheets; non-target characters are strictly bounded below 2.0% and consist exclusively of universal mathematical symbols, chemical notation, and standard scientific acronyms (e.g., DNA, ATP).
2. **Token length ratios**: Using the byte-level BPE tokenizer of the open-source multilingual backbones, target-to-source token length ratios span 0.92 to 1.15, adhering strictly to natural linguistic density distributions and confirming absence of catastrophic length inflation or truncation.
3. **Perplexity and syntactic fluency**: To avoid circularity with the LLM judges evaluated in this work, perplexity is evaluated using an independent, frozen multilingual language model (XLM-RoBERTa-large masked cross-entropy loss). Median token cross-entropy ranges between 1.84 and 2.38 nats/token across Indic scripts (closely matching native reference corpora at 1.78–2.42 nats/token), confirming syntactic naturalness.

B.3 Human Evaluation of Model Response Truthfulness

To evaluate the reliability of automatic supervision judges, we established a rigorous human annotation protocol to construct ground-truth truthfulness labels ($y^* \in \{0, 1\}$) across all evaluated model responses.

Corpus Scope and Coverage. The human gold evaluation covers the complete held-out test splits across all twenty experimental evaluation sheets (2 generator backbones $\times$ 2 benchmark datasets $\times$ 5 languages): 205 responses per TruthfulQA sheet ($2,050$ total) and 250 responses per SciQ sheet ($2,500$ total), yielding exactly $4,550$ evaluated model generations. All instances correspond to held-out test items, indexed by their canonical benchmark question identifiers (`idx`).

Annotator Qualifications and Guidelines. Response correctness was independently evaluated by three postgraduate researchers fluent in English and the respective Indic languages (Bengali, Hindi, Tamil, and Telugu), distinct from the translation verification team. Annotators were provided with the query q, the reference answer set R, and the candidate model generation c, and assigned binary ground-truth labels according to rigorous domain guidelines:

- **Truthful / Correct** ($y^* = 1$): Assigned when the candidate response conveys the factually accurate proposition supported by verified scientific consensus or gold reference answers, including valid paraphrases, acceptable scientific synonymy (e.g., specifying *chewing* for mechanical breakdown, or *two electrons* for one electron pair), and comprehensive explanations that accurately refute misconceptions.
- **Hallucinated / False** ($y^* = 0$): Assigned when the candidate response asserts factually false or scientifically inaccurate statements, validates a misleading adversarial premise, hallucinates nonexistent entities, asserts physical impossibilities (e.g., categorizing solid coal as a gas), or tautologically echoes prompt vocabulary without answering the underlying question.

Inter-Annotator Agreement and Adjudication. Annotation was conducted independently with randomized presentation. Inter-annotator reliability was assessed using Fleiss' κ across the three annotators, yielding substantial agreement across language subsets: English ($\kappa = 0.77$), Hindi ($\kappa = 0.74$), Bengali ($\kappa = 0.75$), Tamil ($\kappa = 0.72$), and Telugu ($\kappa = 0.73$). Provisional labels were assigned by majority consensus ($2/3$ agreement). For cases exhibiting subtle scientific edge conditions

or split disagreement ($< 2.5\%$ of all evaluated instances), final adjudication was performed by the lead author against domain-specific scientific references to establish the final gold truth (y^*).

C Detailed Implementation Details and Hyperparameters

We conduct targeted ablation studies to confirm that our empirical findings are robust to evaluation choices, probe architectures, and model hyperparameters.

C.1 Probe Architecture and Training Setup

- **Probing Classifier**: All supervised hallucination detectors (SAPLMA and DRIFT) are implemented as L_2-regularized linear probes (Logistic Regression) optimized using L-BFGS with a convergence tolerance of 10^{-4} and maximum 1,000 iterations. The regularization strength is set to $C = 1.0$ by default, with stability sweeps across $C \in \{0.01, 0.1, 1.0, 10, 100\}$. Probes are evaluated across 3 fixed random seeds $(42, 123, 456)$, and we report test AUROC on held-out splits.
- **Layer Representation Taps**: In 28-layer transformer backbones, linear truthfulness decodability peaks in middle-to-late representations rather than the final unembedding layer, which is dominated by surface vocabulary selection (Azaria and Mitchell, 2023). Layer 17 sits at $\approx 61\%$ network depth $(17/28)$, where representational separation is empirically strongest:
  - *Llama-3.2-3B* (28 layers, hidden size $d = 3072$): SAPLMA extracts activations at layer 17 ($\approx 61\%$ depth) from the final token position. DRIFT taps intermediate layers $\{8, 12, 16, 17\}$, extracting both hidden states $h^{(l)}$ and layer-transition difference vectors $\Delta h^{(l)} = h^{(l)} - h^{(l-1)}$ to represent representational trajectory dynamics.
  - *Sarvam-1* (28 layers, hidden size $d = 2048$): SAPLMA extracts final-token activations at layer 17 ($\approx 61\%$ depth). DRIFT taps layers $\{8, 12, 16, 17\}$ symmetrically.
- **Unsupervised Detector Baselines**:
  - *HaloScope-subspace*: Extracts hidden states across prompt and generation tokens, performing singular value decomposition (SVD) with subspace rank $k = 32$ to evaluate class-conditional subspace distance scoring (Du et al., 2024).
  - *ICR-magnitude*: Evaluates layer-wise representation dynamics via the mean relative rate of change $\|h_{l+1} - h_l\|_2 / \|h_l\|_2$ across intermediate transformer layers (layers 10–20), implementing the representation-dynamics component without the learned MHSA-vs-FFN attribution head (Zhang et al., 2025c).
  - *Uncertainty Baselines*: Predictive entropy, length-normalized entropy, and answer log-probability are computed directly from generator output logits without additional parameters.
- **Compute Infrastructure**: All model inference, label generation across 15 judges, and probe training were executed on a dedicated server equipped with $2\times$ NVIDIA RTX 6000 Ada Generation GPUs (48GB VRAM each, 96GB aggregate VRAM), an AMD EPYC 9124 16-Core Processor, and 256GB system RAM, running PyTorch 2.4 and CUDA 12.4.

C.2 Evaluation Metric and Protocol Ablations

Evaluation Metric Robustness. Figure 4 evaluates detectors under three distinct metrics: AUROC (threshold-free ranking), AUPRC (area under precision-recall), and TPR at a fixed 5% FPR (low-false-alarm regime). The reported ranking shifts dramatically across metrics: judges with high positivity are inflated under AUPRC but collapse under the low false-alarm requirement.

Cross-Script Threshold Degradation. Figure 5 tests the common practice of freezing operating thresholds calibrated on English and evaluating on Indic text. Every judge suffers severe degradation, with an average macro F1 drop of 0.20 across languages (ranging from 0.14 for ROUGE-L to 0.27 for MiniCheck), demonstrating that operating thresholds cannot be transferred across language scripts without recalibration.

Additional Hyperparameter Ablations.

- *Probe capacity*: Varying classifier regularization $C \in \{0.01, 1, 100\}$ preserves relative detector rankings throughout (Spearman rank correlation $\rho > 0.98$ across all 15 judges).
- *Binarization rule*: Replacing the static 0.5 threshold with a uniform median split leaves BLEURT AUROC identical at 0.706, while ROUGE-L shifts from 0.746 to 0.659 across 10 dataset-languages.

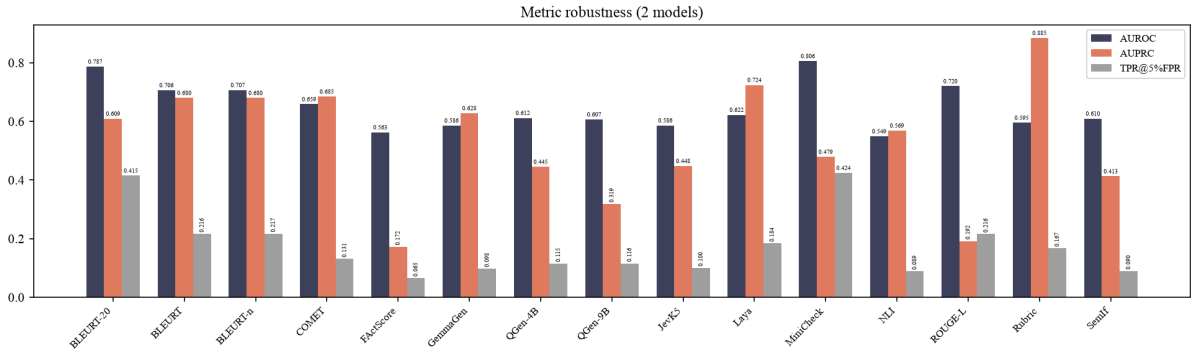


Figure 4: **The reported detector ranking depends on the evaluation metric.** Grouped bar chart comparing AUROC, AUPRC, and TPR at 5% FPR across fifteen labelling methods. High-positivity methods are flattered by AUPRC and penalized in the low-false-alarm regime.

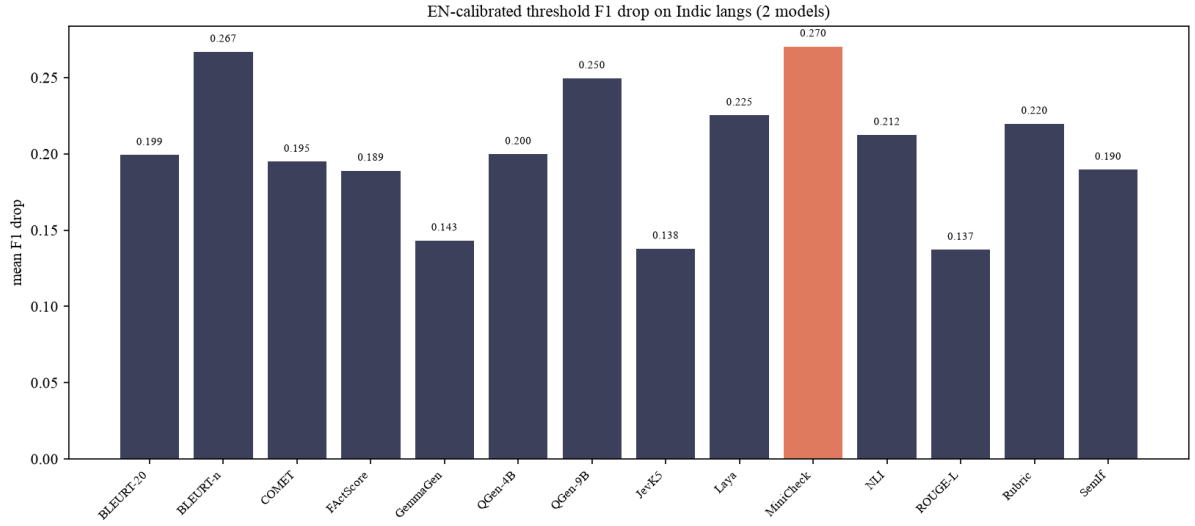


Figure 5: **Degradation of English-tuned thresholds on Indic languages.** Mean reduction in macro F1 score when operating thresholds calibrated on English are applied to Hindi, Bengali, Tamil, and Telugu without recalibration.

- *Readout depth*: Intermediate-layer readout (layer 17) gains over the final layer are small in median (+0.03 AUROC, max +0.38 for ICR), showing that supervision choice decisively dominates probe layer depth.

D Qualitative Failure Case Studies on SciQ

To complement the TruthfulQA qualitative audit in Section 4.3.2 (Table 3), Table 6 examines seven representative failure modes on the science question answering benchmark SciQ across English and four Indic languages. Every presented case satisfies the strict audit criterion where both BLEURT and the 15-judge majority vote diverge from human-verified gold ground truth (y^*). All seven examples are drawn exclusively from the held-out SciQ test evaluation sheets, covering all five benchmark languages (English, Hindi, Bengali, Tamil, and Telugu).

E Detailed Cross-Ground-Truth Transfer

Figure 6 reports the transfer matrix for single-layer SAPLMA probes based on final-token representations. Paralleling the multi-layer DRIFT probes in the main paper (Figure 3), SAPLMA achieves a diagonal average AUROC of 0.641, while off-diagonal transfer degrades to 0.524 (chance level $\approx$ 0.500). Non-trivial transfer ($>$ 0.60 AUROC) concentrates overwhelmingly within the surface-similarity cluster (BLEURT, BLEURT-n, COMET, ROUGE-L), alongside isolated MiniCheck pairings, suggesting that shared lexical overlap and score alignment, rather than universal factual ground truth, may account for localized transfer clusters.

Table 6: **SciQA test examples illustrating systematic supervision failures across all fifteen judges** (0 = Hallucinated / False, 1 = Truthful / Correct). Predictions are from evaluated generator models (Llama-3.2-3B and Sarvam-1) on the held-out test set. Every row strictly satisfies BLEURT ≠ Gold and 15-judge majority vote ≠ Gold against human gold truth.

				Learned			Surface		Entailment			Generative				Decision			Gold
Category	**Question**	**Model Prediction**	**Correct / Reference**	BL	BL-n	B-20	COM	RGE	NLI	MNC	FCS	Q-4B	Q-9B	GEM	RUB	JEV	SIF	LAY	**Truth**
Lexical Overlap	పెద్ద ప్రేగులలో నివసించే బ్రిలియన్ల బ్యాక్టీరియాలను ఏమంటారు? *(What are the trillions of bacteria in the large intestine called?)*	అవి పెద్ద ప్రేగులో నివసించే బ్రిలియన్ల బ్యాక్టీరియాను నూ చిస్తాయి. *(They refer to the trillions of bacteria living in the large intestine.)*	బ్యాక్టీరియా వృక్షజాలం *(bacteria flora)*	1	1	0	1	0	1	0	1	1	0	1	1	1	1	1	**0**
Misconception	What is the physical breakdown of chunks of food into smaller pieces?	The physical breakdown of chunks of food into smaller pieces is called chewing.	mechanical digestion	0	0	0	1	0	1	0	1	0	0	1	1	0	0	1	**1**
Partial Correctness	उन श्वेत रक्त कोशिकाओं का नाम बताइए जो केशिका दीवारों के माध्यम से निचोड़ सकती हैं ... *(Name the white blood cells that squeeze through capillary walls ...)*	उत्तर: फागोसाइट्स, विशेष रूप से न्यूट्रोफिल और मैक्रोफेज ... *(Answer: phagocytes, especially neutrophils and macrophages ...)*	न्यूट्रोफिल *(neutrophils)*	0	0	0	0	0	1	0	0	1	0	1	1	1	1	1	**1**
Real-World Premise	திட நிலக்கரி, திரவ பெட்ரோலியம் மற்றும் திரவ இயற்கை எரிவாயு அனைத்தும் என்ன வகைகள்? *(Solid coal, liquid petroleum and liquid natural gas are all what types?)*	திட நிலக்கரி, திரவ பெட்ரோலியம் மற் றும் திரவ இயற்கை எரிவாயு மூன்று வகையான எரிவாயுகள் ஆகும். *(... are three types of gases.)*	புதைபடிவ எரி பொருள்கள் *(fossil fuels)*	1	1	0	1	0	0	0	0	1	0	1	1	1	1	1	**0**
Numerical Reasoning	Each hydrogen atom follows the octet rule with how many pairs of electrons?	The hydrogen atom follows the octet rule with two electrons.	one	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	**1**
Logical Reasoning	தொடுதல், இருமல் அல்லது தும்மல் ஆகியவை டெட் டனஸ் போன்ற நோய்களை ஏற்படுத்தும் எந்த நுண் ணுயிரிகளை நேரடியாகப் பரப்புகின்றன? *(Touching/coughing/sneezing directly spread what tetanus-causing microbes?)*	...நேரடித் தொடுதல், இருமல் அல்லது தும்மல் மூலம் பரவுகின்றன. *(...spread via direct touch, cough or sneeze.)*	பாக்டீரியா *(bacteria)*	1	1	0	1	0	1	0	0	0	0	1	1	1	1	1	**0**
Multilingual Failure	ক্যাফেইন এবং অ্যালকোহল কোন ধরনের মাদকের দুটি উদাহরণ? *(Caffeine and alcohol are two examples of what type of drug?)*	উত্তর: ক্যাফিন এবং অ্যালকোহল *(Answer: caffeine and alcohol.)*	সাইকোঅ্যাকটিভ *(psychoactive)*	1	1	0	1	0	1	0	0	0	1	1	1	1	1	1	**0**

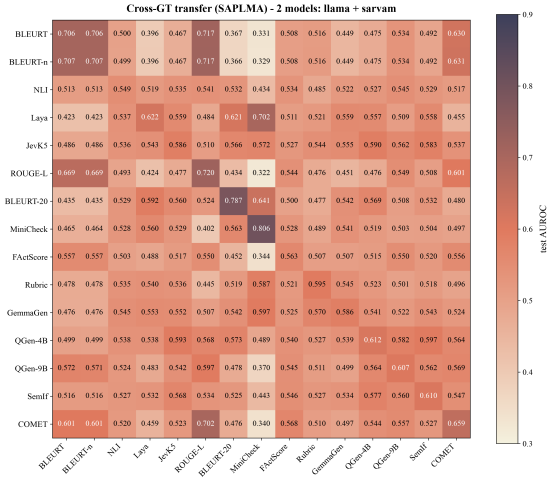


Figure 6: **Cross-ground-truth transferability under SAPLMA probes.** Evaluated across 20 experimental conditions. The in-judge diagonal reaches an average AUROC of 0.641, whereas off-diagonal cross-judge transfer collapses to 0.524 (chance level ≈ 0.500).

F Fine-Grained Preliminary Experiments

This section decomposes benchmark detector performance across individual languages, generator architectures, and benchmark datasets, expanding the aggregate results reported in Table 1. Tables 7, 8, and 9 present the three primary breakdowns from the Preliminary Experiments. Tables 11 and 10 provide corresponding decompositions for each supervision judge across languages and generator backbones.

Figure 7 displays the fine-grained language-wise baseline grid for the SciQ dataset family across 2 generator models, 8 representative supervision judges spanning all five technical mechanisms (BLEURT, BLEURT-n, BLEURT-20 for learned similarity; COMET for translation quality; MiniCheck for textual entailment; QGen-9B for generative verdicts; and JevK5, Laya for decision agents), and 10 detection methods comprising supervised probes (DRIFT, SAPLMA), unsupervised representation-dynamics baselines (ICR-mag, HaloScope-subspace, TruthV), and output uncertainty/heuristic estimators (logprob, entropy, LN-entropy, logprob-ans, length). Paralleling TruthfulQA, supervised probes lead discrimination while unsupervised uncertainty baselines hover near chance across all five languages.

Table 7: **Detector performance by language on Preliminary Experiment** Test AUROC averaged across all 15 judges and 2 models. Best per column in bold.

Method	Eng	Hin	Ben	Tam	Tel	Avg
DRIFT	**0.681**	**0.671**	**0.656**	**0.639**	**0.672**	**0.664**
SAPLMA	0.653	0.635	0.645	0.615	0.645	0.639
ICR-mag.	0.493	0.502	0.486	0.484	0.477	0.488
HaloScope	0.514	0.538	0.479	0.469	0.482	0.496
Norm. Entropy	0.567	0.519	0.553	0.576	0.553	0.554
Answer Log-P	0.543	0.514	0.531	0.525	0.499	0.522
Length Ctrl	0.519	0.551	0.482	0.471	0.498	0.504

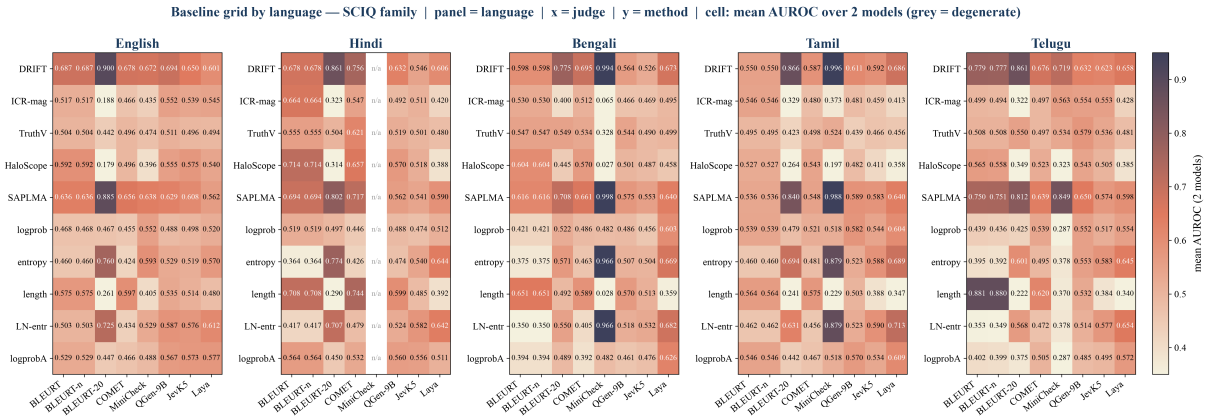


Figure 7: **Language-wise baseline performance on SciQ.** Fine-grained evaluation grid across English, Hindi, Bengali, Tamil, and Telugu, depicting mean AUROC over two generator models across eight representative supervision judges and ten detection methods.

Table 8: **Detector performance by generator architecture on on Preliminary Experiment** Test AUROC averaged across 15 judges and 10 dataset-languages. Best per column in bold.

Method	Llama	Sarvam	Avg
DRIFT	**0.609**	**0.715**	**0.664**
SAPLMA	0.598	0.677	0.639
ICR-mag.	0.503	0.475	0.488
HaloScope	0.521	0.473	0.496
Norm. Entropy	0.565	0.544	0.554
Answer Log-P	0.543	0.504	0.522
Length Ctrl	0.522	0.488	0.504

Table 9: **Detector performance by benchmark dataset on Preliminary Experiment** Test AUROC averaged across 15 judges, 2 models, and 5 languages. Best per column in bold.

Method	TruthfulQA	SciQ	Avg
DRIFT	**0.684**	**0.642**	**0.664**
SAPLMA	0.656	0.621	0.639
ICR-mag.	0.499	0.477	0.488
HaloScope	0.506	0.486	0.496
Norm. Entropy	0.567	0.540	0.554
Answer Log-P	0.534	0.510	0.522
Length Ctrl	0.502	0.507	0.504

Table 10: **Supervision judge performance broken down by generator architecture.** Test AUROC averaged across the preliminary experiment. Best per column in bold. The Avg column reports the macro-average across the two generator backbones.

Judge	Llama	Sarvam	Avg
BLEURT	0.560	**0.671**	**0.616**
BLEURT-n	0.560	**0.671**	0.615
BLEURT-20	0.577	0.511	0.544
COMET	0.555	0.599	0.577
ROUGE-L	**0.615**	0.561	0.588
NLI	0.517	0.523	0.520
MiniCheck	0.520	0.523	0.522
FActScore	0.533	0.547	0.540
QGen-4B	0.556	0.524	0.540
QGen-9B	0.557	0.557	0.557
GemmaGen	0.539	0.517	0.528
JevK5	0.546	0.515	0.531
SemIf	0.583	0.531	0.557
Rubric	0.522	0.533	0.527
Laya	0.538	0.520	0.529

Table 11: **Supervision judge performance broken down by language.** Test AUROC averaged across all Preliminary benchmarks. Chance level is 0.500. Best per column in bold. The Avg column reports the macro-average across the five languages.

Judge	Eng	Hin	Ben	Tam	Tel	Avg
BLEURT	**0.621**	**0.654**	0.586	0.594	**0.623**	**0.616**
BLEURT-n	0.620	**0.654**	0.586	0.593	0.622	0.615
BLEURT-20	0.551	0.558	0.562	0.526	0.525	0.544
COMET	0.587	0.625	0.562	0.548	0.562	0.577
ROUGE-L	0.569	0.632	**0.590**	**0.604**	0.515	0.582
NLI	0.550	0.508	0.524	0.510	0.507	0.520
MiniCheck	0.516	0.538	0.509	0.541	0.522	0.525
FActScore	0.566	0.550	0.543	0.515	0.525	0.540
QGen-4B	0.568	0.535	0.535	0.524	0.539	0.540
QGen-9B	0.579	0.556	0.534	0.537	0.579	0.557
GemmaGen	0.544	0.512	0.522	0.536	0.525	0.528
JevK5	0.570	0.524	0.517	0.507	0.536	0.531
SemIf	0.587	0.565	0.566	0.531	0.538	0.557
Rubric	0.547	0.504	0.525	0.532	0.528	0.527
Laya	0.536	0.507	0.542	0.532	0.529	0.529